Lecture 14 Practice Questions and Solutions

Question 1 (q1_sol.py)

Write a program to help a restaurant cashier calculate the total cost of a meal for a group of people. The program asks the price of each customer's meal, and then prints the total cost of the entire meal. The cashier types "done" when all meals have been entered.

If there are more than 3 customers in the group, then a service charge of 20% is automatically added to the total cost.

All prices printed by the program should be formatted in standard dollar format, i.e. starting with $ and to exactly 2 decimal places.

Sample Run 1

```
Price of meal 1: 10.00
Price of meal 2: 15.99
Price of meal 3: 20.00
Price of meal 4: 25.00
Price of meal 5: done

      Subtotal: $70.00
Service charge: $14.00
         Total: $84.00
```

Sample Run 2

```
Price of meal 1: 6.50
Price of meal 2: 20.00
Price of meal 3: done

      Subtotal: $26.50
         Total: $26.50
```

Sample Solution

```
#
# initialize variables
#
meal_num = 1
subtotal = 0.0

#
# get the prices of the meals
#
meal_price = float(input("Price of meal 1: "))
while meal_price != "done":
```

```
    subtotal += float(meal_price)
    meal_num += 1
    meal_price = input(f"Price of meal {meal_num}: ")

#
# Add a 20% service charge if there are more than 3 customers
#
total = subtotal
service_charge = 0.0
if meal_num > 4: # careful: 4 is correct here, not 3!
    service_charge = subtotal * 0.2
    total += service_charge

#
# print the results
#
print()
print(f"      Subtotal: ${subtotal:.2f}")
if service_charge > 0:
    print(f"Service charge: ${service_charge:.2f}")
print(f"         Total: ${total:.2f}")
```

Question 2 (q2_sol.py)

You have your first job as a software engineer! You work at a children's education software company, and you are in charge of checking if a word contains the letter *m*. The previous developer (who has been promoted to work with *n*) has left you with this function:

```
def has_m_print(word):
    for letter in word:
        if letter == 'm':
            print('this word has an "m"')

# test code
has_m_print('tumble')  # this word has an "m"
has_m_print('Map')     # nothing printed
has_m_print('apple')   # nothing printed
has_m_print('')        # nothing printed
```

Your task is to update the function as follows:

- make it find both lowercase `m` and uppercase `M`
- instead of printing a message, have it return `True` if the word contains an m (either lowercase or uppercase), and `False` otherwise; you should probably change the name of the function
- convert and update the test cases to work with this new function

Sample Solution

```
def has_m(word):
    for letter in word:
        if letter == 'm' or letter == 'M':
            return True
    return False

# test code
print(has_m('tumble'))  # True
print(has_m('Map'))     # True
print(has_m('apple'))   # False
print(has_m(''))        # False
```

Question 3 (q3_sol.py)

Write a program that reads the lines of the file groceries.txt and prints each line on the screen starting with a number.

For example, suppose groceries.txt contains the following:

```
eggs
milk
coffee
more coffee
chocolate
```

Then your program should print:

```
1. eggs
2. milk
3. coffee
4. more coffee
5. chocolate
```

You can assume that the file groceries.txt is in the same directory as your program, and could contain 0 or more lines.

Sample Solution

```
#
# open the file; the 'r' means to open it for reading only
#
input_file = open('empty.txt', 'r')

# print each line on the screen starting with its line number
line_number = 1
for line in input_file:
    print(f"{line_number}. {line.strip()}")
    line_number += 1

#
# close the file
#
input_file.close()
```